Hyperfine structure of the $5d\ ^2D_J$ states in the alkaline-earth Ba ion by fast-ion-beam laser-rf spectroscopy

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(Received 16 September 1985)

The hyperfine structure (hfs) of the $5d\,^2D_{5/2}$ ^{135,137}Ba ions has been measured by collinear fast-beam laser-rf double-resonance spectroscopy, completing the high-precision hfs data for the metastable $^2D_{3/2,5/2}$ doublet. hfs constants, corrected for second-order hfs perturbation, are deduced and compared with the results of many-body perturbation theory. Nuclear quadrupole moments are extracted.

I. INTRODUCTION

Theoretical investigations of the hfs of alkaline-earth ions[1-4] have been recently begun, mainly for three reasons. First, the development of ion-storage and fast-ion-beam techniques has created a source of reliable and high-precision hfs data for ions. Second, the many-body perturbation (MBP) calculations of the first-order, core-polarization, and correlation contributions to the hyperfine fields of these one-electron systems allow a comparison with their counterparts for the isoelectronic alkali-metal atoms. This enlarges the understanding of the trends in the different contributions in both series. Third, nuclear moments extracted from a combination of experimental and MBP data for these ions can reduce the uncertainty and clarify the discrepancies remaining in the deduction of nuclear moments using semiempirical formulas and alkaline-earth atom hfs data.

Whereas the hfs of alkali-metal atoms has been studied extensively by MBP theory, the study of the alkaline-earth ions is far from complete. The only complete investigation for ground and low-lying excited states has been performed for Ca^{+}[1]. For Mg^+ and Ba^+ a complete calculation has been done for the ground state only.[2,3] Very recently, the first-order and core-polarization contributions to the hf fields for low-lying s, p, and d states in all ions of the series have been obtained and the correlation contributions estimated.[4]

Ba^+ is the ion for which most experimental data are available. The experimental hfs of the $5d\,^2D_{5/2}$ ^{135,137}Ba ions presented in this Brief Report complete the high-precision data for the ground and low-lying excited states. A comparison with the MBP results is made and values for the nuclear quadrupole amounts of ^{135,137}Ba ($I=\frac{3}{2}$) are deduced.

II. EXPERIMENTAL PROCEDURE AND RESULTS

The hyperfine structure of $5d\,^2D_{5/2}$ $^{135,137}Ba^+$ has previously been measured by "classical" fast-ion-beam collinear laser spectroscopy (FIBCLS).[5] The analysis of the fluorescent spectra is complicated by the fact that several hyperfine components (residual linewidth ~ 50 MHz) drastically overlap due to the small hyperfine splitting of the $D_{5/2}$ level. We therefore remeasured this hfs by collinear fast-beam laser-rf double-resonance spectroscopy. This recently developed method combines a high sensitivity with a high precision.[6,7]

The apparatus is described in detail in Ref. 7 and the experimental procedure is identical to the reported measurement of the $5d\,^2D_{3/2}$ hfs. We therefore only summarize the main aspects.

A fast beam of metastable $^2D_{5/2}$ ^{135,137}Ba ions, produced with a surface-ionization source and a 11-kV acceleration potential, is sent collinearly through a coaxial rf traveling-wave transmission line. The applied rf field is perpendicular to the beam axis and its frequency can be scanned to induce $F \leftrightarrow F \pm 1$ rf transitions between the hyperfine sublevels of the metastable state. In order to create unequal populations of the F levels and to detect rf resonance transitions, the light of a tunable dye laser is made to interact collinearly with the ion beam in two zones of the beam path, one at each side of the rf zone. These zones consist of open-ended Faraday cages, and resonant laser interaction with one of the hyperfine components of the optical $D_{5/2} \rightarrow P_{3/2}$ (614 nm) transition is achieved by Doppler tuning. By optical depopulation pumping $D_{5/2} \rightarrow P_{3/2} \rightarrow S_{1/2}$, the laser depletes selectively the desired $D_{5/2}$ hfs sublevel F, resulting in a reduction of the fluorescent light detected in the second laser interaction zone. If the frequency-scanned rf field be-

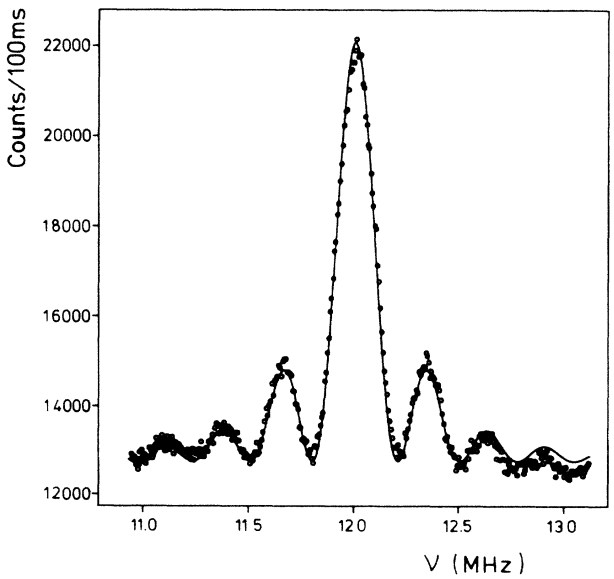


FIG. 1. Spectrum of $5d\,^2D_{5/2}$ $F=3 \leftrightarrow F=4$ resonance in $^{135}Ba^+$ by collinear fast-ion-beam laser-rf double-resonance spectroscopy.

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TABLE I. Experimental hfs splittings in the $5d\,^2D_{5/2}$ $^{135,137}Ba^+$ level.

F-F'	Mass no.	Experimental	+ Corr. 2nd-order hfs perturb.
$\delta\nu_{1\text{-}2}$	135	52.421 (12)	52.423 (12)
(MHz)	137	71.678 (13)	71.681 (13)
$\delta\nu_{2\text{-}3}$	135	49.615 (12)	49.616 (12)
(MHz)	137	62.871 (13)	62.872 (13)
$\delta\nu_{3\text{-}4}$	135	11.990 (13)	11.986 (13)
(MHz)	137	0.49 (6)[a]	0.49 (6)

[a]Calculated from $\delta\nu_{1\text{-}2}$ and $\delta\nu_{2\text{-}3}$, with $C_{5/2}(137)=0$.

comes resonant with a $F \leftrightarrow F \pm 1$ transition, the F level will be repopulated and this is observed as an increase in detected fluorescent light. The rf resonance lines are Doppler shifted and the linewidth is determined by transit time broadening.

As an example, the $D_{5/2}$ $F=3 \leftrightarrow F=4$ rf resonance in $^{135}Ba^+$ is shown in Fig. 1. The full curve through the data points represents the fit according to Ref. 7. The experimental results are summarized in the first column of Table I. These data are already corrected for the small light shifts (<1 kHz) of the hfs levels due to the presence of the detuned laser field in the rf zone.[8]

The quoted errors include the uncertainty stemming from possible shifts due to the presence of reflected rf wave components,[9] which could not be avoided for these low rf frequencies with our present rf equipment. This also explains why the resulting errors are about three times larger than for the higher-frequency $D_{3/2}$ splittings. The $F=3 \leftrightarrow F=4$ splitting in $D_{5/2}$ $^{137}Ba^+$ could not be measured. From the other splittings, a transition frequency of ~ 500 kHz was deduced (see below), which turned out to fall outside the range of our rf equipment.

The experimental hfs coupling constants for the $5d\,^2D_J$ doublet are presented in Table II. For $^{135}Ba^+$, the experimental hfs coupling constants $A_{5/2}$, $B_{5/2}$, and $C_{5/2}$ were deduced from the three hyperfine splittings using the formula of Schwartz.[10] As the octupole constant is negligibly small, just as $C_{3/2}$ (^{135}Ba) and $C_{3/2}$ (^{137}Ba),[7] the $C_{5/2}$ (^{137}Ba) value could safely be taken zero and $A_{5/2}$ and $B_{5/2}$ for ^{137}Ba were deduced from the two measured splittings. Using these hfs coupling constants combined with those for the $D_{3/2}$ level, the hf $D_{5/2}$ splittings were corrected for second-order hfs perturbation by the close-lying $D_{3/2}$ level, according to the formulas given in Ref. 7. The corrected splittings are given in the second column of Table I. The final hfs coupling constants are presented in Table II. The ratios $A_{5/2}(135)/A_{5/2}(137)=0.893(1)$ and $B_{5/2}(135)/B_{5/2}(137)=0.6499(5)$ are in agreement with the accurately measured ratios of nuclear magnetic dipole and electric quadrupole moments[11] [0.893 954 2(48) and 0.649 99(4), respectively], so that up to now the only hyperfine anomaly observed in the ^2D_J doublet is a very small magnetic dipole hyperfine anomaly in the $^2D_{3/2}$ level.[7] In the following, we therefore can restrict our discussion to ^{137}Ba.

III. DISCUSSION

Apart from the complete MBP treatment of the ground-state $S_{1/2}$ hfs of Mg^+, Ca^+, and Ba^+,[1-3] in excellent agreement with experimental A values,[12-14] the only complete MBP calculations for low-lying excited states of alkaline-earth ions have been performed for Ca^+[1] for which no experimental data are available.

The first-order and core-polarization contributions to the A and B/Q factors of low-lying states in the other ions of the series have recently been obtained, and the Ca^+ calculations were used to estimate the correlation contributions for the other ions.[4] This was done by an examination of the trends in the correlation contribution to the hf fields for both the series of alkali-metals and alkaline-earth ions, using both theoretical and experimental data. For the

TABLE II. Experimental hfs coupling constants for the $5d\,^2D_{5/2,3/2}$ $^{135,137}Ba^+$ levels.

	Mass no.	$5d\,^2D_{5/2}$ Experimental	$5d\,^2D_{5/2}$ + Corr. 2nd-order hfs perturb.	$5d\,^2D_{3/2}$[a] Experimental	$5d\,^2D_{3/2}$[a] + Corr. 2nd-order hfs perturb.
A	135	−10.735 (2)	−10.735 (2)	169.5892 (9)	169.5898 (9)
(MHz)	137	−12.028 (11)	−12.028 (11)	189.7288 (6)	189.7296 (7)
B	135	38.688 (10)	38.692 (10)	28.9536 (25)	28.9528 (25)
(MHz)	137	59.528 (43)	59.533 (43)	44.5417 (16)	44.5408 (17)
C	135	−0.000 01 (47)	+ 0.000 01 (47)	0.000 00 (9)	0.000 00 (9)
(MHz)	137	· · ·	· · ·	−0.000 06 (5)	−0.000 06 (5)

[a]Reference 7.

TABLE III. Comparison of theoretical and experimental A factors.

Ion	Level	A_{theor} (MHz) 1st-order + polarization contr.	+ Correlation contr.	A_{expt} (MHz)
$^{137}Ba^+$	$6p\,^2P_{3/2}$	119[a]	126[a]	126.9 (7)[b]
	$5d\,^2D_{3/2}$	150[c]	215	189.7296 (7)
	$5d\,^2D_{5/2}$	−57[c]	−18	−12.028 (11)
$^{87}Sr^+$	$5p\,^2P_{1/2}$	−147[a]	−175[a]	−178.6 (5)[d]

[a]Reference 4.
[b]Reference 5.
[c]Reference 15.
[d]Reference 16.

$5d\,^2D_{3/2,5/2}$ state of Ba^+, MBP values for the first-order and core-polarization contributions to the A and B/Q factors were recently obtained together with an estimate of the correlation contribution to B/Q.[15] This estimate was obtained by putting the ratio of the correlation contribution to the first-order contribution equal to the corresponding ratio that results from the $Ca^+(3d)$ calculation,[1] and it is worthwhile to verify how well this procedure works for $A(5d\,^2D_{3/2,5/2}\ Ba^+)$.

The comparison between MBP results and our experimental data is presented in Tables III and IV. In these tables we include the $6p\,^2P_{3/2}$ Ba^+ state, for which both A and B factors are measured,[5] and also compare the MBP values for the $5p\,^2P_{1/2}$ $^{87}Sr^+$ state with the experimental result[16] which was not included in the compilation of Ref. 4.

A comparison of the third and last column of Table III shows that the MBP results, without correlation contribution, and the experimental A factors do disagree for all states. Inclusion of the correlation contribution, estimated by the described procedure based on the MBP Ca^+ data brings the calculated $P_{1/2,3/2}$ A factors into excellent agreement with the experimental data. Even for the $D_{3/2,5/2}$ levels, a much better agreement between experimental and calculated A factors is obtained.

A direct comparison between the theoretical and our experimental quadrupole interaction data is not possible because nuclear quadrupole moments for Ba, as for most other elements, have not been measured directly. Experimental hfs constants for a number of p-electron states in both atomic and ionic Ba, combined with semiempirical formulas[17] were used to deduce the Q value of ^{137}Ba.[18–21] The mean semiempirical value $Q=0.25(1)$ barn, deduced from our "classical" FIBCLS $P_{3/2}$ data,[5] supports these values for $Q(^{137}Ba)$. The MBP calculations for the $6p\,^2P_{3/2}$ level (see Table IV) resulted in $Q=0.246(2)$ barn, in agreement with the semiempirical result.

In a comparison of the $5d\,^2D$ quadrupole interaction data, it is interesting to consider first the Q value deduced from the $B(5d\,^2D_{3/2,5/2})$ values using the semiempirical formulas. If the radial part of the quadrupole interaction is deduced from the experimental A factors or from the fine-structure splitting, the Q values 0.337 and 0.361 barn, respectively, are obtained. These are in good mutual agreement, as well as consistent with the value $Q=0.34(4)$ barn deduced from other $5d$-electron state hfs data ($6s5d\,^1D,^3D$ Ba I).[22] However, a marked discrepancy between this set of semiempirical Q values deduced from $5d$-electron states and that from $6p$-electron states[23] is revealed, illustrating the shortcomings of the semiempirical procedure. If we look now at the Q values deduced from MBP calculations (Table IV) for the $D_{3/2,5/2}$ levels, we notice excellent agreement with the $Q(P_{3/2})$ values. For completeness, we quote the Q value for ^{135}Ba deduced from the $D_{5/2}$ data: $Q_{135}(D_{5/2})=0.150$ barn. As these Q values are based on incomplete MBP calculations, an uncertainty of about 10% should be taken into account.

The comparison of the experimental hfs constants, presented in this report, with the theoretical data, supports the validity of the estimation of the correlation contribution based on the MBP calculations of the direct and polarization contribution and the complete Ca^+ MBP calculation. It illustrates the capacity of MBP theoretical investigations to describe the hyperfine interaction in low-lying excited states of alkaline-earth ions, although more experimental data and complete MBP calculations for other ions of the series are needed to bring the investigation of the alkaline-earth ions to the level of that of the isoelectronic alkali-metal-atom series.

TABLE IV. Comparison of theoretical and experimental quadrupole interaction constants in $^{13}Ba^+$.

Level	B_{expt} (MHz)	$Q_{semiemp}$ (barn)	B/Q_{theor} (MHz/barn) 1st-order + polarization contr.	+ Correlation contr.	Q (barn)
$6p\,^2P_{3/2}$	92.8 (1.0)[a]	0.25 (1)	341[b]	378[c]	0.246
$5d\,^2D_{3/2}$	44.5408 (17)	0.35 (1)	145[c]	193[c]	0.231
$5d\,^2D_{5/2}$	59.533 (43)	0.35 (1)	196[c]	258[c]	0.231

[a]Reference 5.
[b]Reference 4.
[c]Reference 15.

ACKNOWLEDGMENTS

We wish to thank A.-M. Mårtensson-Pendrill for communication and discussion of results prior to publication. Financial support from the Interuniversitair Instituut voor Kernwetenschappen, Belgium, is acknowledged. Two of the authors (R.E.S., P.D.B.) received support from the Nationaal Fonds voor Wetenschappelijk Onderzoek, Belgium.

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[1]A.-M. Mårtensson-Pendrill and S. Salomonson, Phys. Rev. A **30**, 712 (1984).

[2]S. Ahmad, J. Andriessen, and T. P. Das, Phys. Rev. A **27**, 2790 (1983).

[3]S. Ahmad, J. Andriessen, K. Raghunathan, and T. P. Das, Phys. Rev. A **25**, 2923 (1982).

[4]J.-L. Heully and A.-M. Mårtensson-Pendrill, Phys. Scr. **31**, 169 (1985).

[5]R. E. Silverans, G. Borghs, G. Dumont, and J.-M. Van den Cruyce, Z. Phys. **295**, 311 (1980); M. Van Hove, Ph.D. thesis, Leuven, 1985 (unpublished).

[6]U. Nielsen, O. Poulsen, P. Thorsen, and H. Crosswhite, Phys. Rev. Lett. **51**, 1749 (1983).

[7]M. Van Hove, G. Borghs, P. De Bisschop, and R. E. Silverans, Z. Phys. A **321**, 215 (1985).

[8]G. Borghs, P. De Bisschop, J. Odeurs, R. E. Silverans, and M. Van Hove, Phys. Rev. A **31**, 1434 (1985).

[9]R. E. Silverans, G. Borghs, P. De Bisschop, and M. Van Hove, Phys. Rev. Lett. **55**, 1070 (1985).

[10]C. Schwartz, Phys. Rev. **97**, 380 (1955).

[11]O. Lutz and H. Oehler, Z. Phys. A **288**, 11 (1978).

[12]W. M. Itano and D. J. Wineland, Phys. Rev. A **24**, 1364 (1981).

[13]F. M. Kelly, H. Kuhn, and A. Pery, Proc. Phys. Soc. London, Sect. A **67**, 450 (1954).

[14]R. Blatt, and G. Werth, Phys. Rev. A **25**, 1476 (1982).

[15]A.-M. Mårtensson-Pendrill (private communication).

[16]G. Borghs, P. De Bisschop, M. Van Hove, and R. E. Silverans, Hyperfine Interact. **16**, 177 (1983); P. De Bisschop, Ph.D. thesis, Leuven, 1985 (unpublished).

[17]A. Rosén, Phys. Scr. **8**, 159 (1973).

[18]W. Becker, W. Fischer, and H. Hühnermann, Z. Phys. **216**, 142 (1968).

[19]I. J. Ma and G. zu Putlitz, Z. Phys. A **277**, 107 (1976).

[20]P. E. G. Baird *et al.*, Proc. R. Soc. London, Ser. A **365**, 567 (1979).

[21]A. C. Mueller *et al.*, Nucl. Phys. **A403**, 234 (1983).

[22]M. Gustavsson, G. Olsson, and A. Rosén, Z. Phys. A **290**, 231 (1979).

[23]A recent compilation is given by P. Grundevik *et al.*, Z. Phys. A **306**, 195 (1982).